JMC 2026 Journées de la Matière Condensée
Le plus grand congrès de physique de la matière condensée en France
26-30 octobre 2026
Université de Toulouse
SFP Société Française de Physique

https://jmc2026.sciencesconf.org/

MMQ04: Quantum Embedding Lens on Strongly Correlated Electrons

Symmetry-resolved two-site entanglement as a key to Mott and pseudogap physics in plaquette CDMFT

Gabriele Bellomia[1,2], Carlos Mejuto-Zaera[3], Massimo Capone[2] and Adriano Amaricci[4]

1 - Institute of Solid State Physics, Technischen Universität Wien, Vienna, Austria

2 - International School for Advanced Studies (SISSA), Trieste, Italy

3 - Laboratoire de Physique Théorique, CNRS, Université de Toulouse, Toulouse, France

4 - Istituto Officina dei Materiali, Consiglio Nazionale delle Ricerche (CNR), Trieste, Italy

The strongly correlated electron systems community approaches the third quantum revolution under a two-fold, bold assumption: strong inter-particle interactions grant richer quantum resources than what can be found in other platforms and these resources are operationally accessible. We are thus compelled to carefully assess these aspects, with appropriate measures of entanglement.

I will show how to identify the special entanglement properties of strongly-correlated electrons by symmetry-resolving the **different quantum amplitudes** connecting two sites in the lattice: one contribution is associated to antiferromagnetic fluctuations (**spin singlets**), another accounts for the entanglement induced by the **hopping**, and the last term describes **holon-doublon binding**.

The antiferromagnetic and holon-doublon contributions are directly connected to the charge and parity **superselection rules**, which have been repeatedly shown to faithfully identify the portion of correlations that can be harnessed in realistic quantum information protocols.

I will present **cluster dynamical mean-field theory** (CDMFT) results for salient landmarks in the paramagnetic phase diagram of the two-dimensional Hubbard model at zero temperature. The symmetry resolution of the two-site entanglement allows for a clear characterization of strong quantum correlations, both in the half-filled **Mott-Hubbard insulator** and in the **pseudogap metal** found at small hole-doping fractions. Antiferromagnetic spin-singlet pairs dominate the entanglement content of both the Mott-Hubbard insulator and the under-doped metal, at large U, while a significant holon-doublon entanglement is manifested only in the intermediate coupling regime, where the competition between U and t drives (full or momentum-selective) Mott localization. Remarkably, both terms are suppressed by orders of magnitude in the Fermi liquid regimes found at weak interaction strengths or large doping fractions, whereas the hopping entanglement remains large. Finally, I will briefly showcase a recent **experimental verification** that spin-singlet (or charge superselected) entanglement is confined to nearest neighbors in the pseudogap phase of cold-atom simulations of the Hubbard model, and that it dies not only by overdoping the system, but also when crossing the temperature at which the pseudogap closes.

These findings highlight the strong connection between superselection rules and the special quantum resources brought in by inter-particle interactions.